National Sun Yat-Sen University

SOPC 設計實務及 FPGA 系統整合設計

Homework #2
Due April 3 2026 9:00AM

1. **<Verilog Design>** Design and implement a VMM (***vector-matrix multiplication***) ***accelerator*** as an IP circuit. The design must meet the following requirements:

   - Your design must include at least 16 multiplier units.
   - The supported vector and matrix sizes must be configurable via input signals rather than fixed at design time. However, the accelerator only needs to support a limited set of predefined sizes.
   - You may assume that all input vectors and matrices are stored in on-chip memory, and that the computation results are also written back to on-chip memory. The relevant design requirements are as follows:

   In addition to your Verilog implementation, you must submit a report that includes the following:

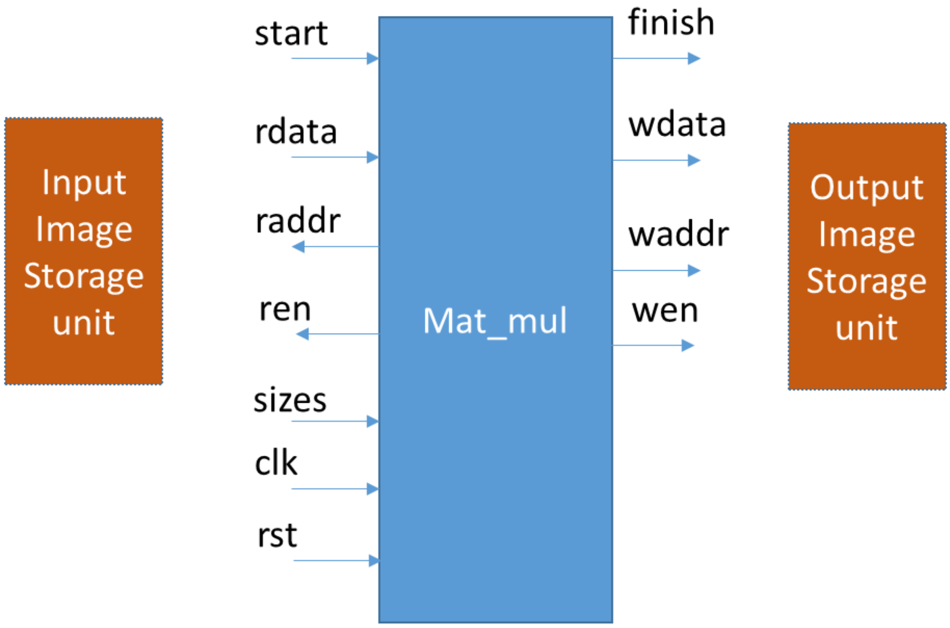
   (a) A synthesis report generated using Vivado Design Suite.

   (b) A detailed description of the I/O interface of your module.
      i. You may follow or adapt the interface shown in the provided example figure.
      ii. Clearly explain how to operate your design.
      iii. Include a timing diagram to illustrate the control flow and usage of your module.

   (c) A detailed circuit diagram of your design.

   (d) A description of the key features of your circuit and an explanation of how the overall architecture operates.

   (e) An analysis of hardware utilization, with particular emphasis on how the multiplier units are used and scheduled in your design.


start
rdata
raddr
ren
sizes
clk
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Input Image Storage unit
Mat_mul
finish
wdata
waddr
wen
Output Image Storage unit